

Prompt Engineering

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Week 2: How LLMs Work (A Non-Technical Primer)

Weekly Learning Objectives

- Explain an LLM as a next-word prediction engine using simple analogies.
- Define "tokenization" and its role in how LLMs process text.
- Describe the basic idea behind the Transformer architecture (self-attention).
- Distinguish between the capabilities and inherent limitations of LLMs.

Key Concepts

- Large Language Model (LLM)
- Transformer Architecture & Self-Attention
- Tokenization & Word Embeddings
- Training vs. Fine-Tuning
- Capabilities & Limitations
- Hallucination

The Core Idea: A Sophisticated Autocomplete

- Think of how your phone suggests the next word as you type—LLMs work on the same principle, just vastly more sophisticated.
- Instead of predicting just words, LLMs predict "tokens" (we'll explain this soon).
- They're trained by reading trillions of words from books, websites, code, scientific papers, and more.
- Through this reading, the model discovers patterns: grammar rules, common facts, writing styles, even cultural context.
- What's remarkable: complex behaviors like answering questions or writing essays emerge from this simple "predict the next piece" mechanism.
- **Key Insight:** No one explicitly programs these rules—the AI learns them by finding patterns in massive amounts of text.

The Scale of Modern LLMs: How Big Are They?

- **Parameters:** Think of these as the "knobs" the AI adjusts during learning. Modern LLMs have billions or even trillions of these adjustable values.
- **To put this in perspective:**
 - GPT-3: 175 billion parameters (like 175 billion adjustable settings)
 - GPT-4: Over 1 trillion parameters
 - LLaMA 2 70B: 70 billion parameters
- **Training Data:** These models read hundreds of billions of text pieces during training—imagine reading millions of books.
- **Cost:** Training cutting-edge models can cost millions of dollars in computing power and electricity.
- Generally, bigger models perform better, though there are limits to this trend.

Small vs. Large Models: What's the Difference?

- Think of it like comparing a high school student to a college professor—both can answer questions, but with different depth.
- **Small Models (fewer parameters):**
 - Faster and cheaper to run
 - Good for simple tasks (basic Q&A, simple summaries)
 - May miss nuances or make more mistakes
 - Example response: "Paris is the capital of France. It has many tourists."
- **Large Models (more parameters):**
 - Better at complex reasoning and nuanced understanding
 - Can handle specialized topics and longer contexts
 - More expensive and slower
 - Example response: "Paris, France's capital, is renowned for its art, culture, and history. The Eiffel Tower, Louvre Museum, and Notre-Dame Cathedral attract millions of visitors annually..."

From Words to Numbers: Tokenization

- Here's a crucial point: computers can't actually "read" text—they only understand numbers.
- **Tokenization** is how we bridge this gap: breaking text into small chunks called "tokens."
- A token might be:
 - A complete word (like "hello")
 - Part of a word (like "un" and "believable")
 - Even a single character or punctuation mark
- **Example:** The phrase "Tokenization is important" might be split into: ["Token", "ization", " is", " important"].
- Each token is then converted to a number the computer can process.

Why Break Words Into Pieces?

- **The Problem:** If we treated every whole word as a token, we'd need millions of different tokens (every word in every language!).
- **The Solution:** Modern LLMs use 32,000-100,000 "sub-word" pieces that can be combined flexibly.
- **Why This Works:**
 - Can handle new or rare words by combining familiar pieces (like building with LEGO blocks)
 - More efficient for the computer to process
 - Works especially well for languages like Turkish or Finnish where words can be very long
- Different AI models may split text differently—like different people cutting a cake into different slice sizes.
- This is why the same text might be slightly different lengths for different AI systems.

From Tokens to Meaning: Embeddings

- After splitting text into tokens, each token becomes a list of numbers called an **embedding**.
- Think of it like GPS coordinates: just as latitude and longitude locate a place on Earth, embeddings locate words in "meaning space."
- Each token gets represented by 768 to 12,288 numbers that capture its meaning and relationships to other words.
- **The Magic:** Words with similar meanings end up close together in this mathematical space.
- **Famous Example:** The relationship between "king" and "queen" is mathematically similar to the relationship between "man" and "woman."
- The AI learns these relationships from data—no human explicitly teaches it that kings and queens are related!

Imagine a Map of Word Meanings

- If we could visualize this "meaning space," related words would appear in neighborhoods together:
 - Animal neighborhood: cat, dog, horse, elephant
 - Color district: red, blue, green, yellow
 - Action zone: run, walk, jump, sprint
- **Context Matters:** Modern LLMs are smart—they understand that "bank" means something different in:
 - "I walked along the river bank" (shoreline)
 - "I deposited money at the bank" (financial institution)
- The same word gets different embeddings based on context!

The Brain of the LLM: The Transformer

- In 2017, researchers published a breakthrough paper titled "Attention Is All You Need" that changed everything.
- The key innovation: **self-attention**—a way for AI to understand which words are important for understanding other words.
- **Think about this sentence:** "The cat sat on the mat, and **it** was happy."
- You instantly know "it" refers to "the cat," not "the mat." How? You're paying attention to the relationship between words.
- Self-attention lets the AI do the same thing: mathematically determine which words should influence the understanding of other words.
- This replaced older approaches that read text one word at a time, making modern AI both faster and smarter.

How Self-Attention Works (Simplified)

- Imagine reading a sentence where each word asks: "Which other words should I pay attention to?"
- For each word, the system creates three pieces of information:
 - **Query:** "What am I looking for?"
 - **Key:** "What information do I contain?"
 - **Value:** "What information can I provide?"
- The system compares all the Queries and Keys to calculate **attention scores**—essentially asking "how relevant is each word to each other word?"
- These scores determine how much each word influences the understanding of every other word.
- **The Result:** Each word gets a rich, context-aware representation that incorporates information from the entire sentence.

Multi-Head Attention: Many Perspectives at Once

- Here's where it gets even more interesting: the AI doesn't use just one attention mechanism—it uses many in parallel.
- **Example:** GPT-3 uses 96 different "attention heads" in each layer, and has 96 layers! That's over 9,000 attention mechanisms working together.
- Think of it like having multiple experts examining the same text:
 - One expert focuses on grammar (subject-verb agreement, sentence structure)
 - Another on meaning (synonyms, related concepts)
 - Another on long-distance connections (what "it" refers to paragraphs earlier)
- All these perspectives get combined to create a rich, nuanced understanding of the text.

The Transformer Architecture: Putting It All Together

- **Step 1 - Input:** Text is converted to embeddings (those number lists we discussed)
- **Step 2 - Position Information:** The system adds information about word order (first word, second word, etc.)
- **Step 3 - Processing Layers:** The text flows through many layers (12 to 96+ in modern models)
 - Each layer applies multi-head attention (understanding word relationships)
 - Then applies additional processing to refine understanding
- **Step 4 - Output:** Finally, the system produces probabilities for what token should come next
- Think of it as multiple stages of increasingly sophisticated reading comprehension.

The Training Process: How Do They Learn?

- **Pre-training:** The AI reads massive amounts of text—books, websites, articles, code, and more.
- **The Learning Game:** For each piece of text, try to predict the next word. Get it wrong? Adjust those billions of parameters slightly. Repeat trillions of times.
- It's like learning a language by reading millions of books and constantly testing yourself on what word comes next.
- **Time & Resources:** This process takes weeks to months, running on thousands of powerful computers working together.
- **The Result:** A "foundation model"—an AI with broad knowledge about language, facts, reasoning patterns, and more.

Fine-Tuning: Making Them Useful and Safe

- After basic training, the AI isn't ready for public use—it needs refinement.
- **Fine-Tuning:** Additional training on carefully selected examples to specialize the model's behavior.
- **Instruction Tuning:** Teaching the AI to follow instructions by showing it many examples of questions and good answers.
- **Learning from Human Feedback:**
 - Humans review and rank different AI responses
 - The AI learns to produce responses that humans find helpful, harmless, and honest
 - This helps align the AI with human values and expectations
- Systems like ChatGPT, Claude, and Gemini all go through extensive fine-tuning before public release.

Capabilities and Limitations

What LLMs Are Remarkably Good At:

LLMs are remarkably flexible—they can write, summarize, translate, analyze, code, and more without being explicitly programmed for each task. This general-purpose adaptability is revolutionary.

Key Limitations to Keep in Mind:

- Hallucination: can produce plausible but incorrect or fabricated facts.
- Biases: reflect stereotypes and biases present in training data.
- No-true understanding: statistical pattern learners, not conscious reasoners.
- Outdated/incomplete knowledge: training cutoff and dataset gaps.
- Context-window limits: cannot attend to arbitrarily long documents.
- Sensitivity to phrasing: small prompt changes produce different outputs.
- Privacy & safety risks: may leak training artifacts or be misused.

Emergent Abilities: Unexpected Superpowers

- Something fascinating happens as models get bigger: new abilities suddenly appear that smaller models didn't have.
- **Examples of emergent abilities:**
 - Learning from just a few examples (like humans do)
 - Explaining step-by-step reasoning
 - Following complex, multi-part instructions
 - Solving problems they've never seen before
- Here's what's remarkable: nobody explicitly programmed these abilities—they spontaneously emerged when the models reached a certain size.
- **Open Question:** Are these truly new abilities, or are we just measuring things differently?
- Understanding this helps us anticipate what even larger future models might be capable of.

Activity 1: Breaking Down Sentences Into Tokens

Let's experiment with tokenization! We'll take several sentences and first predict how an AI might break them into tokens. Then we'll use an online tokenizer tool (platform.openai.com/tokenizer) to see the actual breakdown. How close were our predictions? What surprises us?

Activity 2: Spot the Hallucination

Time to be AI detectives! We'll examine AI-generated paragraphs that contain subtle factual errors. Using fact-checking and search engines, we'll work together to identify which information is hallucinated and discuss why the AI might have made these mistakes.

Discussion Questions

- If an LLM is just predicting the next word, why does it feel like it understands our questions?
- Now that we know LLMs can hallucinate, how should this change the way we use them for academic research?
- What dangers might arise if people aren't aware of the biases embedded in LLMs?
- Does the concept of "emergent abilities" challenge our understanding of what intelligence really is?
- If LLMs can only "read" limited amounts of text at once, what creative strategies could we use to analyze very long documents?
- Should we trust AI-generated content differently than human-written content? Why or why not?

Key Takeaways

- LLMs are sophisticated pattern-matching systems that learn from enormous amounts of text—think of them as "autocomplete on steroids."
- Tokens are the building blocks of text (like LEGO bricks), and understanding tokenization helps explain why AI sometimes behaves unexpectedly.
- The Transformer architecture's self-attention mechanism is what enables them to understand context and relationships between words.
- Bigger models generally perform better, but there are trade-offs in cost, speed, and complexity.
- LLMs are powerful tools, but they have real limitations—hallucinations, biases, and knowledge gaps—that we must be aware of.
- Context windows limit how much text an LLM can "see" at once—this matters for long documents.
- Understanding how LLMs work helps us write better prompts and use AI more effectively in our own fields.

Assignment for Next Week

Short Essay (500 words)

Topic: "Explain LLMs to a High School Student Using an Analogy"

Requirements:

- Choose one central analogy that helps explain how LLMs work
- Explain at least 3 key concepts: tokens, training, and one limitation
- Make it accessible—no jargon without explanation
- Be creative! The best analogies make complex ideas simple

Due Date: Next class session

Thank You!

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Questions?